

The Mind Stone

MIRAGE 2.0 — Reverse Engineering — Official Writeup

CATEGORY	PORT	FILES
Reverse Engineering / Self-Modifying Code	1711/tcp	mind_stone_final (statically linked, stripped)
VULNERABILITY	STATUS	
N/A — JIT-decoded validator reversal	Solved	

Challenge Premise

A “Vision Core” recovery terminal that claims to be a “SELF-MODIFYING CORE”. The menu offers: [1] Inspect telemetry, [2] Attempt stone recovery, [3] Disconnect. Option [1] prints flavor text including a long hex “TRACE” string that turns out to be an unused red herring — it plays no role in any decode path.

Reversing the Binary

Option [2] requires an input of exactly 17 characters (`strlen(buf) == 0x11`). Two anti-analysis gates run before any real validation:

- **Anti-debug:** reads `/proc/self/status`; a non-zero `TracerPid` diverts to a decoy path.
- **Anti-timing:** a 90,000-iteration LCG busy loop bracketed by two `CLOCK_MONOTONIC_RAW` reads; elapsed time over ~20ms (consistent with single-stepping) also diverts to the same decoy path.

The decoy path prints a plausible-looking, plaintext, hardcoded flag baked directly into `.rodata` — visible to strings, but never reachable on a genuine run:

```
MIRAGE{th3_m1nd_st0n3_b3l0ngs_t0_th4n0s}  <-- decoy, only shown if traced
```

On the real path, the binary `mmaps` a fresh RW page, decodes a 349-byte embedded blob into it using a rolling keystream (`state = 0xd15ea5e5`, then `state = state * 0x19660d + 0x3c6ef35f` each step, plus a byte offset of `+0x3d` per iteration), then `mprotects` the page executable and **calls into it**, passing a pointer to the user's input. The decoded machine code turns out to be 17 small chained blocks, one per input byte:

```
movzx al, [rdi]
inc   rdi
xor   al, X_i
add   al, Y_i
cmp   al, Z_i
jne   fail
```

Each position's (`X`, `Y`, `Z`) constants were extracted from the decoded blocks and inverted algebraically: `input[i] = X_i XOR ((Z_i - Y_i) & 0xFF)`, recovering the required input purely from static decode — no brute force needed.

Participant Solve Steps

```
$ nc <host> 1711
VISION@CORE> 2
[INPUT] Stone recovery value: MINDSTONE-TRUE-42
```

FLAG

```
MIRAGE{v1s10n_f0und_th3_m1nd_st0n3!_42}
```

Verification

Confirmed against a live, natively-run instance of the service (not attached to a debugger, to avoid tripping the anti-debug gate): sending MINDSTONE-TRUE-42 produced “[+] FINAL SIGNAL ACCEPTED” followed by the flag above, matching the offline static decode exactly and confirming the process is fully deterministic (no randomness involved in either the keystream or the check).